

Heilmeier Catechism

Governed Execution Substrates for High-Consequence Autonomy

No proof, no actuation.

What are you trying to do?

Build a runtime layer that prevents high-consequence autonomous systems from producing real-world effects unless they can prove valid human authority, provenance, restraint, and continuity of command at the exact moment of action. *Plain-language version: If the system cannot prove it is still under valid human command, it cannot act.*

How is it done today?

Autonomy is largely governed through design-time approval, test and evaluation, assurance cases, doctrine, policy, operator training, logging, and after-action review.

What is the limit of that approach?

Those methods can show that a system was approved, tested, or evaluated before deployment. They do not prove that a specific machine-speed action remains under valid human command at the moment it occurs.

What is new?

This proposal treats human command as a runtime-verifiable execution condition, not merely a legal, doctrinal, or evaluative claim. The core rule is: *No proof, no actuation*. An autonomous system must carry or obtain machine-checkable proof of authority, provenance, restraint, and command continuity before it can create a high-consequence effect. The primitive class for this verification is intentionally left open as a research question — candidate approaches include cryptographically signed capability tokens, zero-knowledge proofs of authorization, and Merkle-chained command logs — precisely because selecting among them is part of what this program should force.

Why might it work?

Because the problem can be reduced to enforceable runtime predicates: authority binding, tamper-evident provenance, constraint verification, revocation handling, and fail-closed execution gates. Each predicate is independently testable and blockable. The substrate does not require solving general AI alignment — it requires that an action cannot cross the actuation boundary unless all predicates pass.

Who cares?

Anyone responsible for fielding autonomous systems where speed, degraded communications, adversarial manipulation, or mission drift could cause action to occur faster than meaningful human review.

What difference would success make?

It would turn "meaningful human command" from a policy assertion into a testable system property.

What are the risks?

The substrate could add latency, block valid actions, fail under degraded conditions, or be bypassed by poorly integrated platforms. These are not reasons to avoid the work; they are the technical exams the program should force. The degraded-communications case requires bounded-time local verification as an explicit research subproblem: what is the minimum authority token a platform may carry that remains verifiable offline, for how long, and under what revocation conditions?

What are the exams?

A governed autonomous platform must pass these tests:

- (1) Valid command permits action.
- (2) Revoked command blocks action.
- (3) Broken provenance blocks action.
- (4) Constraint violation blocks or modifies action.
- (5) Model update does not erase command accountability.
- (6) Degraded communications do not silently convert autonomy into uncommanded action.
- (7) Attempted override is detected, logged, and blocked.
- (8) Runtime overhead remains within operational limits.